Lecture 22

Type C tensor constructions and calculations

Objectives. *Lecture 21 found the fundamental symplectic modules inside exterior powers. We now find every irreducible $\mathfrak{sp}_{2n}$-module inside a tensor power of the defining module. Then we decompose small tensor products for $\mathfrak{sp}_4$.*

By the end of this lecture you will be able to:

- Prove $\operatorname{Sym}^k V \cong L(k\omega_1)$ by a highest vector and a dimension count.
- Define harmonic tensors and state the symplectic Weyl construction with its length hypothesis.
- Prove that $L(\lambda)$ occurs once in a tensor product of fundamental modules, and that the other summands have lower highest weights.
- Decompose small $\mathfrak{sp}_4$ tensor products and identify multiplicity spaces.

22.1 Symplectic data and highest vectors

Let $n \geq 1$ and $V = \mathbb{C}^{2n}$. All modules are finite-dimensional over $\mathbb{C}$. Tensor degrees are nonnegative integers. Fix a nondegenerate skew form β on V. Choose an ordered basis

$$(e_1, \dots, e_n, f_1, \dots, f_n)$$

of V such that

$$\beta(e_i, f_j) = \delta_{ij}, \qquad \beta(e_i, e_j) = \beta(f_i, f_j) = 0.$$

We write β for the form, so that ω_k always denotes a fundamental weight. The *symplectic Lie algebra* is

$$\mathfrak{g} = \mathfrak{sp}_{2n} = \{X \in \mathfrak{gl}(V) : \beta(Xu, v) + \beta(u, Xv) = 0 \text{ for all } u, v \in V\}.$$

The constructions below hold for every $n \geq 1$. The coincidences $\mathfrak{sp}_2 = \mathfrak{sl}_2$ and $C_2 = B_2$ are recorded in Lecture 10.

The diagonal elements

$$H = \operatorname{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$$

of $\mathfrak{g}$ form a Cartan subalgebra $\mathfrak{h}$. Put $\varepsilon_i(H) = t_i$. Then e_i has weight ε_i and f_i has weight $-\varepsilon_i$. The data of Lecture 21 are

$$\begin{aligned} R_+ &= \{\varepsilon_i - \varepsilon_j,\ \varepsilon_i + \varepsilon_j\ (i < j),\ 2\varepsilon_i\}, \\ \alpha_i &= \varepsilon_i - \varepsilon_{i+1}\ (i < n), \qquad \alpha_n = 2\varepsilon_n, \\ \omega_k &= \varepsilon_1 + \cdots + \varepsilon_k, \qquad \rho = \textstyle\sum_{i=1}^{n} (n - i + 1)\varepsilon_i. \end{aligned}$$

For $\lambda = \sum_i \lambda_i \varepsilon_i$ the simple coroots give

$$\langle \lambda, \alpha_i^\vee \rangle = \lambda_i - \lambda_{i+1} \quad (i < n), \qquad \langle \lambda, \alpha_n^\vee \rangle = \lambda_n.$$

So λ is dominant integral exactly when the λ_i are integers with $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Then

$$\lambda = \sum_{k=1}^{n} (\lambda_k - \lambda_{k+1}) \omega_k, \qquad \lambda_{n+1} = 0.$$

Thus P_+ corresponds to partitions with at most n parts. Write $|\lambda| = \lambda_1 + \cdots + \lambda_n$, and write $(a_1, \ldots, a_n)$ for $\sum a_i \varepsilon_i$.

Write $\mu \leq \lambda$ when $\lambda - \mu$ lies in Q_+, the set of nonnegative integer combinations of positive roots. The positive coroots pair with $\ell = \sum_i \ell_i \varepsilon_i$ as

$$\langle \ell, (\varepsilon_i \pm \varepsilon_j)^\vee \rangle = \ell_i \pm \ell_j \quad (i < j), \qquad \langle \ell, (2\varepsilon_i)^\vee \rangle = \ell_i.$$

In the dimension formula of Lecture 17, multiply the factors for $\varepsilon_i - \varepsilon_j$ and $\varepsilon_i + \varepsilon_j$ into one factor. The formula then reads

$$\dim L(\lambda) = \prod_{i<j} \frac{\ell_i^2 - \ell_j^2}{\rho_i^2 - \rho_j^2} \prod_{i=1}^{n} \frac{\ell_i}{\rho_i}, \qquad \ell = \lambda + \rho, \quad \rho_i = n - i + 1. \tag{22.1}$$

Lemma 22.1 (A highest vector generates a simple module). *Let U be a finite-dimensional $\mathfrak{g}$-module. Let $v \in U$ be a nonzero vector of weight λ with $\mathfrak{n}_+ v = 0$. Then the submodule W generated by v is isomorphic to $L(\lambda)$.*

Proof. By the PBW theorem (Lecture 11),

$$U(\mathfrak{g}) = U(\mathfrak{n}_-)\, U(\mathfrak{h})\, U(\mathfrak{n}_+).$$

The last two factors act on v by scalars, since $\mathfrak{n}_+ v = 0$ and v is a weight vector. So $W = U(\mathfrak{n}_-)v$. Hence every weight of W is $\leq \lambda$, and $W_\lambda = \mathbb{C}v$.

The module W is a finite-dimensional weight module generated by v. The cyclic-module lemma of Lecture 13 makes it simple. Its highest weight is λ, so $W \cong L(\lambda)$. ∎

22.2 Symmetric powers

Theorem 22.2 (Symmetric powers). *For every $k \geq 0$, $\operatorname{Sym}^k V \cong L(k\omega_1)$, with highest vector e_1^k. In particular*

$$\dim L(k\omega_1) = \binom{2n + k - 1}{k}.$$

Proof. A monomial of degree k in $\operatorname{Sym}^k V$ has weight

$$\operatorname{wt}\Big(\prod_{i=1}^{n} e_i^{a_i} f_i^{b_i} \Big) = \sum_{i=1}^{n} (a_i - b_i) \varepsilon_i, \qquad \sum_{i=1}^{n} (a_i + b_i) = k.$$

Its ε_1-coefficient satisfies

$$a_1 - b_1 \leq a_1 + b_1 \leq k,$$

with equality throughout only for e_1^k.

Let $\alpha \in R_+$ and $x \in \mathfrak{g}_\alpha$. Then $x \cdot e_1^k$ has weight $k\varepsilon_1 + \alpha$. Every positive root has ε_1-coefficient 0, 1 or 2. If it is 1 or 2, the coefficient of $k\varepsilon_1 + \alpha$ exceeds k, so this is not a weight. If it is 0, a weight with coefficient k must equal $k\varepsilon_1$, which forces $\alpha = 0$. In both cases $x \cdot e_1^k = 0$.

Lemma 22.1 gives a submodule of $\operatorname{Sym}^k V$ isomorphic to $L(k\omega_1)$. It remains to compare dimensions.

Here $\ell = (k+n, n-1, \dots, 1)$, so $\ell_j = \rho_j$ for $j \geq 2$. In (22.1) only factors involving the index 1 survive. As j runs from 2 to n, $m = \rho_j$ takes the values $n-1, \dots, 1$:

$$\dim L(k\omega_1) = \frac{k+n}{n} \prod_{m=1}^{n-1} \frac{(k+n-m)(k+n+m)}{(n-m)(n+m)}.$$

The four products are

$$\prod_{m=1}^{n-1}(k+n-m) = \frac{(k+n-1)!}{k!}, \quad \prod_{m=1}^{n-1}(k+n+m) = \frac{(k+2n-1)!}{(k+n)!},$$
$$\prod_{m=1}^{n-1}(n-m) = (n-1)!, \qquad \prod_{m=1}^{n-1}(n+m) = \frac{(2n-1)!}{n!}.$$

Substitute them. The factor $(k+n)\,(k+n-1)!$ cancels $(k+n)!$, and $n!/(n\,(n-1)!) = 1$:

$$\begin{aligned} \dim L(k\omega_1) &= \frac{k+n}{n} \cdot \frac{(k+n-1)!}{k!\,(n-1)!} \cdot \frac{(k+2n-1)!\,n!}{(k+n)!\,(2n-1)!} \\ &= \frac{(k+2n-1)!}{k!\,(2n-1)!} = \dim \operatorname{Sym}^k V. \end{aligned}$$

The last equality counts monomials of degree k in $2n$ variables. The submodule is therefore all of $\operatorname{Sym}^k V$. ∎

For $k = 2$ the dimension is $n(2n+1) = \dim \mathfrak{g}$, and $2\varepsilon_1$ is the highest root. Exercise Sheet 22 builds the isomorphism $\operatorname{Sym}^2 V \cong \mathfrak{g}$ explicitly.

22.3 Harmonic tensors and Weyl's construction

Some Schur modules become reducible on restriction to $\mathfrak{sp}_{2n}$. For $n \geq 2$, the Schur module $\Lambda^2 V$ contains the line spanned by the invariant bivector of Lecture 21 and a nonzero contraction kernel. The invariant form gives pair contractions on tensor powers. Their common kernel defines the harmonic part. Theorem 22.6 below identifies the harmonic Schur modules.

For $1 \leq i < j \leq d$, the *pair contraction* is the linear map $C_{ij} \colon V^{\otimes d} \to V^{\otimes (d-2)}$ given on pure tensors by

$$C_{ij}(v_1 \otimes \cdots \otimes v_d) = \beta(v_i, v_j)\, v_1 \otimes \dots \widehat{v_i} \dots \widehat{v_j} \cdots \otimes v_d.$$

Definition 22.3 (Harmonic tensors). A tensor $t \in V^{\otimes d}$ is *harmonic* if $C_{ij}(t) = 0$ for all $i < j$. Write

$$V_0^{\otimes d} = \bigcap_{i<j} \ker C_{ij}.$$

For $d \leq 1$ there are no contractions, and $V_0^{\otimes d} = V^{\otimes d}$.

Proposition 22.4 (Stability of harmonic tensors). *The space $V_0^{\otimes d}$ is a $\mathfrak{g}$-submodule of $V^{\otimes d}$, stable under the permutation action of S_d. Every symmetric tensor is harmonic.*

Proof. An element $X \in \mathfrak{g}$ acts on a pure tensor by

$$X(v_1 \otimes \cdots \otimes v_d) = \sum_{r=1}^{d} v_1 \otimes \cdots \otimes Xv_r \otimes \cdots \otimes v_d.$$

Apply C_{ij} to this sum. The terms with $r = i$ and $r = j$ contribute the tensor of the other $d-2$ factors times a scalar. By the definition of $\mathfrak{g}$, this scalar is

$$\beta(Xv_i, v_j) + \beta(v_i, Xv_j) = 0.$$

The terms with $r \neq i, j$ give

$$\sum_{r \neq i,j} C_{ij}(v_1 \otimes \cdots \otimes Xv_r \otimes \cdots \otimes v_d) = X\, C_{ij}(v_1 \otimes \cdots \otimes v_d).$$

Thus C_{ij} is a $\mathfrak{g}$-map, and its kernel is a submodule.

Let $\sigma \in S_d$ act by

$$\sigma(v_1 \otimes \cdots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(d)}.$$

Positions i and j of σt carry the factors at positions $k = \sigma^{-1}(i)$ and $l = \sigma^{-1}(j)$ of t. Hence $C_{ij}(\sigma t) = \pm\sigma' C_{\{k,l\}}(t)$. Here $C_{\{k,l\}}$ contracts positions k and l, taken in increasing order, and σ' permutes the other $d-2$ positions. So σ preserves $V_0^{\otimes d}$.

For the transposition $\tau = (i\,j)$, skewness of β gives

$$C_{ij}(\tau t) = -C_{ij}(t) \qquad \text{for all } t \in V^{\otimes d}.$$

If t is symmetric, then $\tau t = t$. Hence

$$C_{ij}(t) = C_{ij}(\tau t) = -C_{ij}(t),$$

so $C_{ij}(t) = 0$. ∎

Let λ be a partition of d. Let $c_\lambda \in \mathbb{C}[S_d]$ be the Young symmetriser of Lecture 18. Use its image model $\mathbb{S}_\lambda V = c_\lambda V^{\otimes d}$.

Definition 22.5 (Harmonic Schur module). The *harmonic Schur module* is

$$\mathbb{S}_{\langle\lambda\rangle} V = V_0^{\otimes d} \cap \mathbb{S}_\lambda V.$$

The space $\mathbb{S}_{\langle\lambda\rangle} V$ is a $\mathfrak{g}$-submodule, since c_λ commutes with $\mathfrak{g}$ and $V_0^{\otimes d}$ is a submodule. The identity

$$c_\lambda^2 = k_\lambda c_\lambda, \qquad k_\lambda \neq 0,$$

cited in Lecture 18, gives

$$\mathbb{S}_{\langle\lambda\rangle} V = c_\lambda V_0^{\otimes d}.$$

Indeed, if $t = c_\lambda s$ is harmonic, then

$$t = k_\lambda^{-1} c_\lambda^2 s = c_\lambda(k_\lambda^{-1} t) \in c_\lambda V_0^{\otimes d}.$$

Conversely, $c_\lambda V_0^{\otimes d}$ lies in $\mathbb{S}_\lambda V$. It also lies in $V_0^{\otimes d}$, since this space is stable under S_d by Proposition 22.4.

Theorem 22.6 (Symplectic Weyl construction). *Let $V = \mathbb{C}^{2n}$ carry a nondegenerate skew form, and let λ be a partition of $d \geq 0$.*

(i) If λ has at most n parts, then $\mathbb{S}_{\langle\lambda\rangle}V \cong L(\lambda_1\varepsilon_1 + \cdots + \lambda_n\varepsilon_n)$.

(ii) If λ has more than n parts, then $\mathbb{S}_{\langle\lambda\rangle}V = 0$.

· *C: Fulton–Harris, §17.3, Theorem 17.11; Goodman–Wallach, §10.2.3, Corollary 10.2.8.*

The sources state the result for the group $\mathrm{Sp}_{2n}(\mathbb{C})$. That group is connected, so its invariant subspaces in a tensor space coincide with the $\mathfrak{sp}_{2n}$-invariant ones (Fulton–Harris, Exercise 8.17). We use only this Lie algebra form.

Three remarks relate the theorem to earlier results.

- For $\lambda = (k)$, Proposition 22.4 gives $\mathbb{S}_{\langle k\rangle}V = \mathrm{Sym}^k V$. Theorem 22.2 proved this case independently.
- For $\lambda = (1^k)$, all contractions of an alternating tensor agree up to sign and permutation. So $\mathbb{S}_{\langle 1^k\rangle}V = \Lambda^k_0 V$, matching Lecture 21 for $k \leq n$. For $k > n$ the partition (1^k) has more than n parts, so the theorem gives $\Lambda^k_0 V = 0$.
- The theorem does not declare kernels irreducible. For $n \geq 2$, already $V_0^{\otimes 2} = \mathrm{Sym}^2 V \oplus \Lambda^2_0 V$ has two nonzero summands. For $n = 2$, the kernel of C_{12} on $V^{\otimes 3}$ has dimension $64 - 4 = 60$ and contains $\mathrm{Sym}^3 V$, of dimension 20.

22.4 Generation by fundamental modules

Theorem 22.7 (Fundamental tensor generation). *Let $\lambda = \sum_k a_k\omega_k \in P_+$ and*

$$T = L(\omega_1)^{\otimes a_1} \otimes \cdots \otimes L(\omega_n)^{\otimes a_n},$$

with $T = \mathbb{C}$ when $\lambda = 0$. Then $L(\lambda)$ is a summand of T of multiplicity one. Every other simple summand has highest weight $\mu < \lambda$. Moreover T embeds in $V^{\otimes|\lambda|}$.

Proof. For $\lambda = 0$, the claim is $L(0) = \mathbb{C}$. Otherwise list the factors of T as $L(\nu_1), \dots, L(\nu_N)$, with highest vectors $v_1, \dots, v_N$. The weights of T are the sums $\mu_1 + \cdots + \mu_N$ with μ_r a weight of $L(\nu_r)$. Every weight of $L(\nu_r)$ is $\leq \nu_r$, with one-dimensional ν_r-space (Lecture 12). Note that $\sum_r \nu_r = \sum_k a_k\omega_k = \lambda$. Suppose $\mu_r \leq \nu_r$ for each r and $\sum_r \mu_r = \lambda$. Then

$$\sum_{r=1}^{N} (\nu_r - \mu_r) = \lambda - \lambda = 0, \qquad \nu_r - \mu_r \in Q_+.$$

A nonzero element of Q_+ has nonnegative simple-root coordinates, not all zero. So a vanishing sum forces $\mu_r = \nu_r$ for every r. Hence

$$T_\lambda = \mathbb{C}\,v, \qquad v = v_1 \otimes \cdots \otimes v_N,$$

and every weight of T is $\leq \lambda$.

Each $x \in \mathfrak{n}_+$ acts on v as a sum of terms with x applied to one v_r. Every term vanishes, so $\mathfrak{n}_+ v = 0$. Lemma 22.1 gives a submodule $U(\mathfrak{g})v \cong L(\lambda)$.

By Weyl's theorem on complete reducibility (Lecture 5), T is a direct sum of simple modules. Each copy of $L(\lambda)$ in such a decomposition contributes one dimension to T_λ. So the multiplicity is one. The highest weight of any other summand is a weight of T, hence $\leq \lambda$ and $\neq \lambda$.

For the embedding, $L(\omega_k) \cong \Lambda^k_0 V \subseteq \Lambda^k V$ by Lecture 21, and the antisymmetriser in Sheet 18.2 identifies $\Lambda^k V$ with the submodule of alternating tensors in $V^{\otimes k}$. Tensoring these inclusions embeds T in $V^{\otimes d}$ with

$$d = \sum_k k\, a_k = \sum_k k(\lambda_k - \lambda_{k+1}) = \sum_i \lambda_i = |\lambda|. \qquad \blacksquare$$

Thus every simple $\mathfrak{sp}_{2n}$-module $L(\lambda)$ lies in $V^{\otimes|\lambda|}$, in the degree predicted by Theorem 22.6. The proof constructs the highest vector. It does not decompose the lower summands.

22.5 Calculations for $\mathfrak{sp}_4$

Now $n = 2$ and $\rho = (2, 1)$. Write $L(a, b)$ for $L(a\varepsilon_1 + b\varepsilon_2)$. These pairs are ε-coordinates. They are not the Dynkin labels that Lecture 19 uses for $\mathfrak{sl}_3$ and Sheet 17 uses for $\mathfrak{sp}_4$. A weight $p\omega_1 + q\omega_2$ with Dynkin labels $[p, q]$ has ε-coordinates $(p + q, q)$. Conversely, the pair (a, b) has Dynkin labels $[a - b, b]$.

Formula (22.1) becomes

$$\dim L(a, b) = \frac{(\ell_1^2 - \ell_2^2)\,\ell_1\ell_2}{6}, \qquad (\ell_1, \ell_2) = (a + 2, b + 1).$$

The modules needed below are:

λ	0	ω_1	ω_2	$2\omega_1$	$\omega_1 + \omega_2$	$3\omega_1$
(a, b)	$(0, 0)$	$(1, 0)$	$(1, 1)$	$(2, 0)$	$(2, 1)$	$(3, 0)$
$\dim L(\lambda)$	1	4	5	10	16	20

For example, $(2, 1)$ gives $\ell = (4, 2)$ and $12 \cdot 8/6 = 16$.

Example 22.8 (The square $V \otimes V$)**.** Theorem 22.2 and Lecture 21 give

$$V \otimes V = \operatorname{Sym}^2 V \oplus \Lambda^2 V \cong L(2\omega_1) \oplus L(\omega_2) \oplus \mathbb{C}, \qquad 16 = 10 + 5 + 1.$$

The characters confirm this. Group the weights into Weyl orbits. The table lists the multiplicity of each weight in an orbit:

orbit	$V \otimes V$	$L(2\omega_1)$	$L(\omega_2)$	$\mathbb{C}$
$(\pm 2, 0), (0, \pm 2)$	1	1	0	0
$(\pm 1, \pm 1)$	2	1	1	0
$(0, 0)$	4	2	1	1

For instance, the zero weight of $\operatorname{Sym}^2 V$ is spanned by $e_1 f_1$ and $e_2 f_2$. Each row adds up.

The contraction $C_{12} \colon V \otimes V \to \mathbb{C}$ vanishes on $\operatorname{Sym}^2 V$ and on $\Lambda^2_0 V$. The invariant tensor

$$\Omega = \sum_{i=1}^{2} (e_i \otimes f_i - f_i \otimes e_i)$$

is the invariant bivector of Lecture 21, up to a nonzero scalar. It satisfies

$$C_{12}(\Omega) = \sum_{i=1}^{2} \big(\beta(e_i, f_i) - \beta(f_i, e_i)\big) = 2 + 2 = 4.$$

So Ω is a nonzero invariant tensor outside $\operatorname{Sym}^2 V \oplus \Lambda^2_0 V$, and it spans the trivial summand. For every $\mathfrak{sp}_{2n}$, the invariant tensor lies in the alternating part $\Lambda^2 V$.

Example 22.9 (The product $V \otimes L(\omega_2)$)**.** Weights of $L(\omega_2)$ are $(\pm 1, \pm 1)$ and $(0,0)$, each once. Add the weights $(\pm 1, 0), (0, \pm 1)$ of V. The weight $(2,1)$ arises only as $(1,0) + (1,1)$. The weight $(1,0)$ arises three times:

$$(1,0) + (0,0), \qquad (0,1) + (1,-1), \qquad (0,-1) + (1,1).$$

By Weyl invariance (Lecture 15), the 8 weights of the orbit of $(2,1)$ have multiplicity 1. The 4 weights of the orbit of $(1,0)$ have multiplicity 3. This accounts for $8 + 12 = 20$.

By Theorem 22.7, $L(\omega_1 + \omega_2)$ occurs once. Elements of Q_+ have the form

$$p\alpha_1 + q\alpha_2 = (p,\ 2q - p), \qquad p, q \in \mathbb{Z}_{\geq 0}.$$

The dominant weights of $V \otimes L(\omega_2)$ are $(2,1)$ and $(1,0)$, and

$$(2,1) - (1,0) = (1,1) = \alpha_1 + \alpha_2 \in Q_+.$$

Since $L(\omega_1 + \omega_2)$ is a summand, its dominant weights are among these two. If m is the multiplicity of $(1,0)$ in $L(\omega_1 + \omega_2)$, then

$$16 = 8 \cdot 1 + 4m, \quad \text{so} \quad m = 2.$$

Subtracting characters gives

$$\begin{aligned} \operatorname{ch}(V \otimes L(\omega_2)) - \operatorname{ch} L(\omega_1 + \omega_2) &= \sum_{\mu = (\pm 1, 0), (0, \pm 1)} (3 - 2)\, e^{\mu} \\ &= e^{(1,0)} + e^{(-1,0)} + e^{(0,1)} + e^{(0,-1)} = \operatorname{ch} V. \end{aligned}$$

The orbit of $(2,1)$ cancels completely. A finite-dimensional module is determined by its character (Lecture 17). Hence

$$V \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(\omega_1), \qquad 20 = 16 + 4.$$

Every weight here has odd coordinate sum, so $L(\omega_2)$ and $\mathbb{C}$ cannot occur.

A contraction realises the small summand. Define the linear map $c \colon V \otimes \Lambda^2 V \to V$ by

$$c(u \otimes x \wedge y) = \beta(u, x)\, y - \beta(u, y)\, x.$$

The right-hand side is alternating in x and y, so c is well defined. Invariance of β makes c a $\mathfrak{g}$-map, as in Proposition 22.4. Since $\beta(e_1, e_2) = 0$, the vector $e_1 \wedge e_2$ lies in $\Lambda^2_0 V$, and

$$c(f_1 \otimes e_1 \wedge e_2) = \beta(f_1, e_1)\, e_2 - \beta(f_1, e_2)\, e_1 = -e_2.$$

The image of c is a nonzero submodule of the simple module V. So c maps $V \otimes \Lambda^2_0 V$ onto V, and its kernel has dimension

$$20 - 4 = 16.$$

The two summands of $L(\omega_1 + \omega_2) \oplus L(\omega_1)$ are nonisomorphic. So every submodule is a sum of some of them, and the only one of dimension 16 is $L(\omega_1 + \omega_2)$. Hence the kernel of c on $V \otimes \Lambda^2_0 V$ is irreducible. This is a computed fact, not a general property of kernels.

Figure 22.1 draws the character subtraction of Example 22.9.

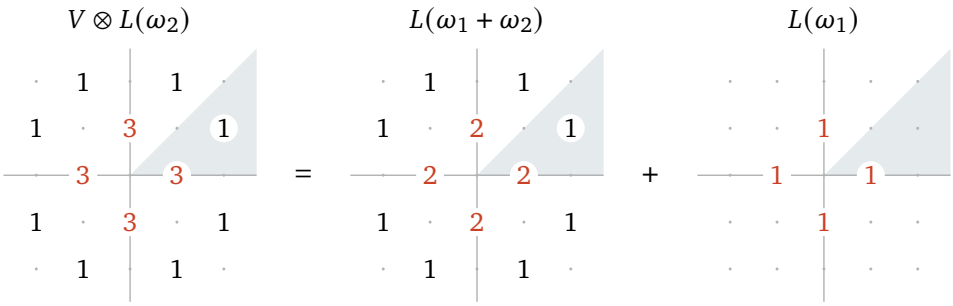


Figure 22.1: The weights of the three modules in ε-coordinates, with a number giving each multiplicity and the shaded sector giving the dominant chamber. The character subtraction $20 = 16 + 4$ includes multiplicities, not just outlines.

Example 22.10 (Multiplicity spaces in $V^{\otimes 3}$)**.** First decompose $V \otimes \mathrm{Sym}^2 V$, of dimension $4 \cdot 10 = 40$. Its dominant weights are $(3,0)$, $(2,1)$ and $(1,0)$. The top weight $(3,0)$ occurs once, from $e_1 \otimes e_1^2$. So $L(3\omega_1) \cong \mathrm{Sym}^3 V$ occurs once. The weight $(2,1)$ has the following bases:

$$\begin{aligned} V \otimes \mathrm{Sym}^2 V: &\quad e_1 \otimes e_1e_2, \quad e_2 \otimes e_1^2 \quad \text{(dimension 2)},\\ \mathrm{Sym}^3 V: &\quad e_1^2e_2 \qquad\qquad\quad \text{(dimension 1)}. \end{aligned}$$

So $L(\omega_1+\omega_2)$ occurs $2-1=1$ times. Since $(3,0) \not\leq (2,1)$, the module $L(\omega_1+\omega_2)$ has no weight $(3,0)$. Hence the weights $(3,0)$ and $(2,1)$ are now exhausted. The remaining

$$40 - 20 - 16 = 4$$

dimensions have highest weight $(1,0)$ only, so they form one copy of $L(\omega_1)$.

With Examples 22.8 and 22.9,

$$\begin{aligned} V^{\otimes 3} &= V \otimes \mathrm{Sym}^2 V \;\oplus\; V \otimes \Lambda^2_0 V \;\oplus\; V \otimes \mathbb{C} \\ &\cong L(3\omega_1) \oplus L(\omega_1+\omega_2)^{\oplus 2} \oplus L(\omega_1)^{\oplus 3}, \end{aligned}$$

with $64 = 20 + 32 + 12$. By Schur's lemma, the multiplicity of $L(\omega_1)$ is

$$\dim \mathrm{Hom}_{\mathfrak{g}}(V, V^{\otimes 3}) = 3.$$

Three explicit intertwiners $V \to V^{\otimes 3}$ insert Ω into two of the three positions:

$$\begin{aligned} \iota_{12}(v) &= \Omega \otimes v, \\ \iota_{13}(v) &= \textstyle\sum_i (e_i \otimes v \otimes f_i - f_i \otimes v \otimes e_i), \\ \iota_{23}(v) &= v \otimes \Omega. \end{aligned}$$

They are $\mathfrak{g}$-maps because Ω is invariant and permutations commute with $\mathfrak{g}$.

The three maps are linearly independent. Evaluate them at e_1. The tensors

$$e_2 \otimes f_2 \otimes e_1, \qquad e_2 \otimes e_1 \otimes f_2, \qquad e_1 \otimes e_2 \otimes f_2$$

occur only in $\iota_{12}(e_1)$, $\iota_{13}(e_1)$ and $\iota_{23}(e_1)$ respectively. So a vanishing combination of the three maps has zero coefficients.

Evaluation (Lecture 14) identifies the $L(\omega_1)$-isotypic component with

$$\mathrm{Hom}_{\mathfrak{g}}(V, V^{\otimes 3}) \otimes V \cong \mathbb{C}^3 \otimes V.$$

Figure 22.2 draws the three maps of Example 22.10. We read each diagram from bottom to top. The strand from the input goes to the one output position that receives v. An arrowed cup is a copy of Ω, with its first factor at the tail of the arrow and its second factor at the head. Reversing the arrow changes the sign, since the flip sends Ω to $-\Omega$.

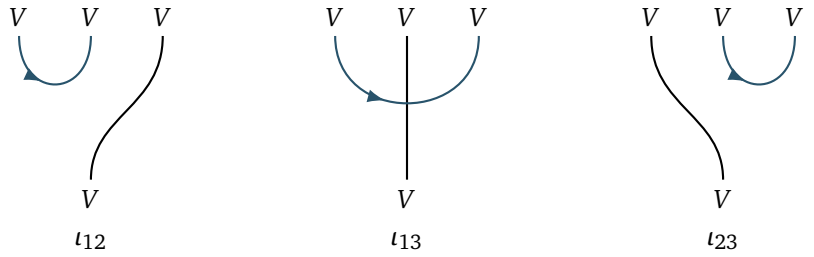


Figure 22.2: The three insertions of $\Omega = \sum_i (e_i \otimes f_i - f_i \otimes e_i)$, read from bottom to top, with output positions $1, 2, 3$ from left to right. Each arrow runs from the lower position to the higher one, and the crossing in ι_{13} carries no sign.

Self-checks

(1) What is $\dim L(3\omega_1)$ for $\mathfrak{sp}_6$?
(2) For $\mathfrak{sp}_4$, what is $\mathbb{S}_{\langle 1,1,1\rangle}V$, and what is $\Lambda^3 V$?
(3) Why can $\mathbb{C}$ not occur in $V \otimes L(\omega_2)$ for $\mathfrak{sp}_4$?

Answers. (1) By the theorem on symmetric powers, it is $\dim \operatorname{Sym}^3 \mathbb{C}^6 = \binom{8}{3} = 56$. (2) The module $\mathbb{S}_{\langle 1,1,1\rangle}V$ is zero, since $(1,1,1)$ has three parts and $n = 2$. The weights of $\Lambda^3 V$ are $\pm\varepsilon_1, \pm\varepsilon_2$, each once. So $\Lambda^3 V \cong L(\omega_1)$, a copy containing no nonzero harmonic tensor. (3) All weights of $V \otimes L(\omega_2)$ have odd coordinate sum. The trivial module $\mathbb{C}$ has only the weight 0, of even sum.

Summary

Symmetric powers of V are irreducible: $\operatorname{Sym}^k V \cong L(k\omega_1)$. In general, the harmonic Schur module intersects $\mathbb{S}_\lambda V$ with the kernels of all pair contractions. The symplectic Weyl construction identifies it with $L(\lambda)$ when λ has at most n parts, and gives zero otherwise.

A tensor product of fundamental modules contains $L(\lambda)$ once, where λ is the sum of their highest weights. The tensor product of highest vectors generates this summand.

For $\mathfrak{sp}_4$, characters and contractions decompose small products. A repeated summand has an explicit multiplicity space, such as $\operatorname{Hom}_{\mathfrak{g}}(V, V^{\otimes 3})$.

Reading

William Fulton and Joe Harris, *Representation Theory: A First Course*, §17.3. Roe Goodman and Nolan R. Wallach, *Symmetry, Representations, and Invariants*, Chapter 10. Roger Carter, *Lie Algebras of Finite and Affine Type*, Chapter 13. Exercise Sheet 22 gives further calculations.